

Name: _____

Student ID: _____

Grade: _____

Full Exam or only Linear Algebra? _____

WARNING: Only the questions of the exam chosen above will be graded.

Question:	1	2	3	4	5	6	7	8	9	10	11	Total
Points:	3	2	1	1	2	6	4	4	5	5	3	36
Score:												

Calculus Grade = #points of questions (1 - 5) + 1 = _____

Linear Algebra Grade = $\frac{\text{\#points of questions (6 - 11)}}{3} + 1 = \underline{\hspace{2cm}}$

If Full Exam chosen, Final Grade = Calculus Grade * 0.3 + Linear Algebra Grade * 0.7

Read the instructions carefully!

- The use of a calculator, book or notes is **not** allowed.
- Answer the questions in the spaces provided. If you run out of room for an answer, continue on the back of the page, but make sure the final answer starts on the front.
- You can ask for extra paper to do your calculations, but you shall not submit it, the graded answers will be the ones on these papers, so make sure the final answers are properly explained here, starting in the blank space provided and finishing on the back if necessary.
- Answers without supporting work will not be given full credit, so make sure to explain the theoretical basis of what you do for full score.
- If you feel like you will be short in time, prioritize questions you know or the ones with higher score count.

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1. **(3 pts.) Calculus:** If R is the region bounded above by the graph of the function $f(x) = 9 - \left(\frac{x}{2}\right)^2$ and below by the graph of the function $g(x) = 6 - x$, find the area of region R .

2. **(2 pts.) Calculus:** Compute the slope of the tangent line to the function $f(x) = \sqrt{1 + \cos(2x)}$ at $x = \frac{\pi}{4}$.

3. **(1 pts.) Calculus:** Consider the following piecewise function:

$$f(x) = \begin{cases} 3x - 2 & x < 2 \\ g(x) & x = 2 \\ x^2 & x > 2 \end{cases}$$

Give a function $g(x)$ such that $f(x)$ is continuous and justify your choice.

4. **(1 pts.) Calculus:** Does $\lim_{x \rightarrow \frac{\pi}{2}} \tan x$ exist? Why?

5. **(2 pts.) Calculus:** Evaluate the following integral

$$\iint_R y + x^2 dA$$

where R is the region $R = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 1, y \geq 0, y \leq x\}$.

6. Let

$$A = \begin{bmatrix} 1 & 2 & 0 & -1 & 3 \\ 0 & 1 & 2 & 1 & 4 \\ 1 & 3 & 2 & 0 & 7 \end{bmatrix}.$$

(a) **(1 pts.)** Without doing any computations, can the system $Ax = 0$ have a unique solution?

(b) **(3 pts.)** Find a basis for $\text{Nul } A$.

(c) **(2 pts.)** Find a basis for $\text{Col } A$.

7. Let $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$ be any square matrix.

(a) **(1 pts.)** Show that if $A^2 - 3A + I = 0$, then $A^{-1} = 3I - A$.

(b) **(2 pts.)** Show that if $a_{ij} = 0$ for $i \neq j$ and $a_{ij} \neq 0$ if $i = j$, then A is invertible.

(c) **(1 pts.)** Under the conditions of b), find the inverse of A .

8. **(4 pts.)** Let the vector space V be the set of all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ and $\mathcal{C}^1(\mathbb{R})$ the subset of V of all the functions that are differentiable. Show that $\mathcal{C}^1(\mathbb{R})$ is a subspace of V .

9. Let A be a square matrix.

- (a) **(1 pts.)** Show that if a matrix A is diagonalizable with diagonal matrix D and change of basis matrix P , then $A^k = PD^kP^{-1}$, where k is a positive integer.

- (b) **(4 pts.)** Let $A = \begin{bmatrix} 1 & -6 \\ 2 & -6 \end{bmatrix}$. Use the information in question a) to find an explicit formula for the entries of A^k in terms of k (you don't need to solve part a) to use the result). Explain carefully every step.

10. Let $w_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ and $w_2 = \begin{bmatrix} 3 \\ 3 \\ -1 \\ -1 \end{bmatrix}$ and $y = \begin{bmatrix} 6 \\ 0 \\ 2 \\ 0 \end{bmatrix}$, and let $W = \text{Span}\{w_1, w_2\}$.

(a) (**3 $\frac{1}{2}$ pts.**) Find a basis for W consisting of two orthogonal vectors.

(b) (**1 $\frac{1}{2}$ pts.**) Express y as the sum of a vector in W and a vector in $W^\perp$.

11. Mark each of the following statements as true or false. If the statement is true, give a proof. If the statement is false, give a proof or provide a counterexample.

(a) **(1 pts.)** Let $T(\mathbf{x}) = (x_2, x_3, x_1^2)$. Since $T(0) = 0$, T is a linear map.

(b) **(1 pts.)** If $V = \mathbb{R}^2$, $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$, and $\mathcal{C} = \{\mathbf{c}_1, \mathbf{c}_2\}$, then row reduction of $[\mathbf{c}_1 \ \mathbf{c}_2 \ \mathbf{b}_1 \ \mathbf{b}_2]$ to $[I \ P]$ produces a matrix P that satisfies $[\mathbf{x}]_{\mathcal{B}} = P[\mathbf{x}]_{\mathcal{C}}$ for all $\mathbf{x}$ in V .

(c) **(1 pts.)** If $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is an orthonormal basis for a subspace W of $\mathbb{R}^n$, then

$$\text{proj}_W \mathbf{y} = (\mathbf{y} \cdot \mathbf{u}_1)\mathbf{u}_1 + \cdots + (\mathbf{y} \cdot \mathbf{u}_p)\mathbf{u}_p$$

